Interdendritic Fluid Flow and Macrosegregation; Influence of Gravity

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An analysis is given of interdendritic flow behavior during solidification of castings and ingots, assuming resistance to flow is as in other types of porous media. Driving forces for the flow are solidification contractions and gravity acting on a fluid of variable density. Detailed flow calculations are given for horizontal, unidirectional, steady-state solidification, using aluminum-copper alloys as examples. Conditions are quantitatively described under which gravity induced convection becomes an important contributory cause of macrosegregation. A critical condition of flow is shown to produce local melting with resulting formation of "channel-type" segregates. Qualitative examples are given of application of the ideas presented to interpretation of macrosegregation in commercial ingots, with specific reference to centerline segregation and "channel-type" segregation, including "V" segregates, "A" segregates and "freckles".

IN a series of recent papers we have examined analytically and experimentally macrosegregation which results from interdendritic fluid flow.[1-4] In that work, details of fluid flow behavior were assumed (without hydrodynamic calculation), and resulting macrosegregation behavior was then calculated. The sole driving force for the assumed flow was liquid-solid shrinkage, and experiments were such that this was the only driving force operating. Solute and/or thermally induced convection were not present in experiments performed since solidification was unidirectional upwards, with liquid density decreasing in the vertical direction.

In this paper we now consider the hydrodynamics of interdendritic fluid flow, as influenced by solidification contraction, liquid contraction during solidification, and gravity acting on a fluid of variable density. We view the liquid-solid region of a casting or ingot as a porous medium in which volume fraction of porosity varies with fraction solid and is continuously changing during solidification. The size of individual pores depends both on fraction solid and microstructure (*e.g.*, dendrite arm spacing). Standard techniques for calculating flow through porous media are applied to this specialized problem, and final macrosegregation determined from the calculated flow distribution. Solutions are limited in this paper to the time-independent case of horizontal, unidirectional, steady-state solidification; comparison of theory with experiment will be considered in a later paper.

It will be seen that the nature and extent of the segregation are strongly influenced, for examples cited herein, by changes in liquid density during solidification. They are also influenced by solidification rate and temperature gradient, phase diagram characteristics, and resistance of the liquid-solid region to interdendritic flow. Conditions necessary for formation of "channel type" segregates will be quantitatively described. These segregates (*e.g.*, "freckles") have previously been shown in two important papers[5,6] to result from interdendritic fluid flow.

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SOLIDIFICATION MODEL

Fig. 1 shows schematically a liquid-solid "mushy" region of a solidifying ingot. Dendrite tips are approximately at the liquidus isotherm. Liquid near the roots of the dendrites is highly segregated, and this is shown by the darker shading. Dendrites themselves are highly schematized. They may be either columnar or equiaxed; in either case, the dendrite arms form a very fine network throughout the mushy zone. As example, a typical width of the mushy zone is of the order of 10 cm. Spacing of dendrite arms in such a zone would be the order of 10^{-2} cm, as shown in the insert of Fig. 1.

The basic model used to describe the above solidification is that employed in previous work.[1,4] A small volume element is considered at coordinates x, y, z within the mushy zone. A suitable element is shown in Fig. 2; it is large enough that fraction solid within it at any time is exactly the local average, but small enough that it can be treated as a differential element during solidification. Solute enters or leaves the element only by liquid flow. Mass flow in or out of the element by diffusion is neglected. Liquid composition and temperature vary only differentially across the element at any time, t. Liquid composition is related to temperature by the liquidus of the appropriate phase diagram. Solute redistribution within this volume element can now be described by the "local solute redistribution equation" given earlier:

$$\frac{\partial g_L}{\partial C_L} = -\left(\frac{1-\beta}{1-k}\right)\left[1 + \frac{\mathbf{v}\cdot\nabla T}{\epsilon}\right]\frac{g_L}{C_L} \qquad [1]$$

where:

g_L = volume fraction liquid

C_L = liquid composition

β = solidification shrinkage

k = equilibrium partition ratio

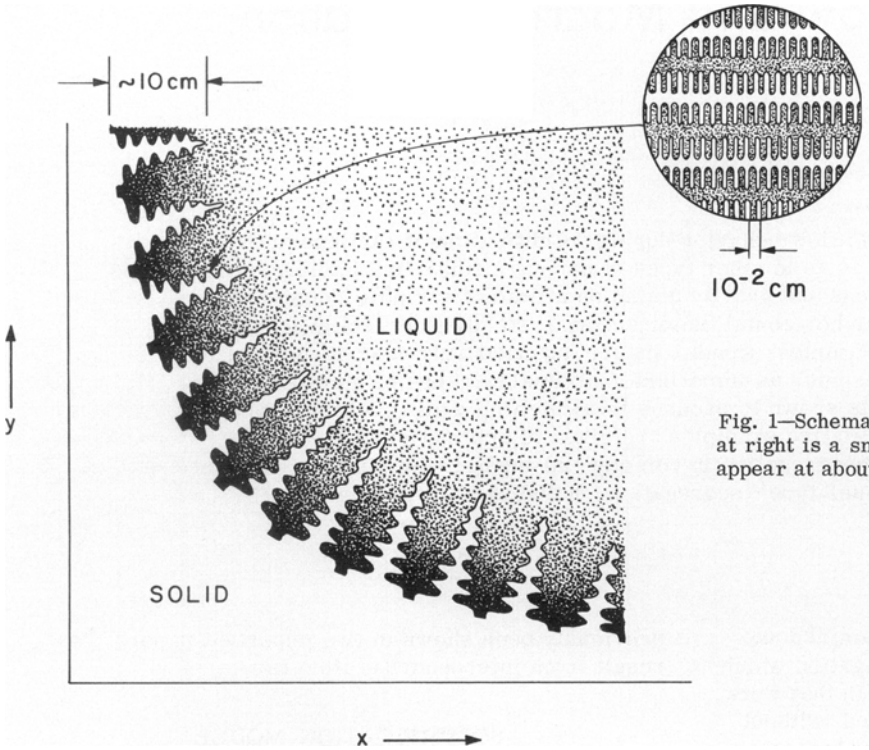

Fig. 1—Schematic illustration of solidification model (Insert at right is a small section of liquid-solid region as it would appear at about 66 times magnification.)

$\mathbf{v}$ = interdendritic flow velocity

T = temperature

ϵ = rate of temperature change

Each of the quantities of Eq. [1] is that obtained in the volume element at x, y, z at time, t, during solidification. In addition to assumptions given above, other assumptions of Eq. [1] are: no diffusion in the solid during solidification, constant solid density, and no pore formation. These assumptions are to be used throughout the remainder of this work.

In this work, as in the previous papers, macrosegregation is calculated by integration of Eq. [1]. The partition ratio, k, and solidification shrinkage, β, are, in general, functions of C_L. ∇T and ϵ are functions of temperature, and hence, of C_L; these quantities are readily measured experimentally or, in principle, calculated from heat flow considerations. Evaluation of $\mathbf{v}$ is discussed below; when this is also known as a function of C_L (or g_L), Eq. [1] is readily integrated.

BASIC FLUID FLOW RELATIONS

A central aim of this paper is to present a method for calculating $\mathbf{v}$. The method given is similar to that employed for fluid flow calculations in other types of porous media.[7-9] The major differences being that liquid composition and volume fraction liquid are allowed to vary with time (as a result of solidification). We assume first that interdendritic flow velocity through the volume element of Fig. 2 is linearly related to pressure gradient; *i.e.*, that D'Arcy's Law applies:*

*Note that D'Arcy's Law here is written for mean interdendritic flow velocity, $\mathbf{V}$, rather than "superficial velocity" as is usually the case. "Superficial velocity" is $g_L\mathbf{V}$.

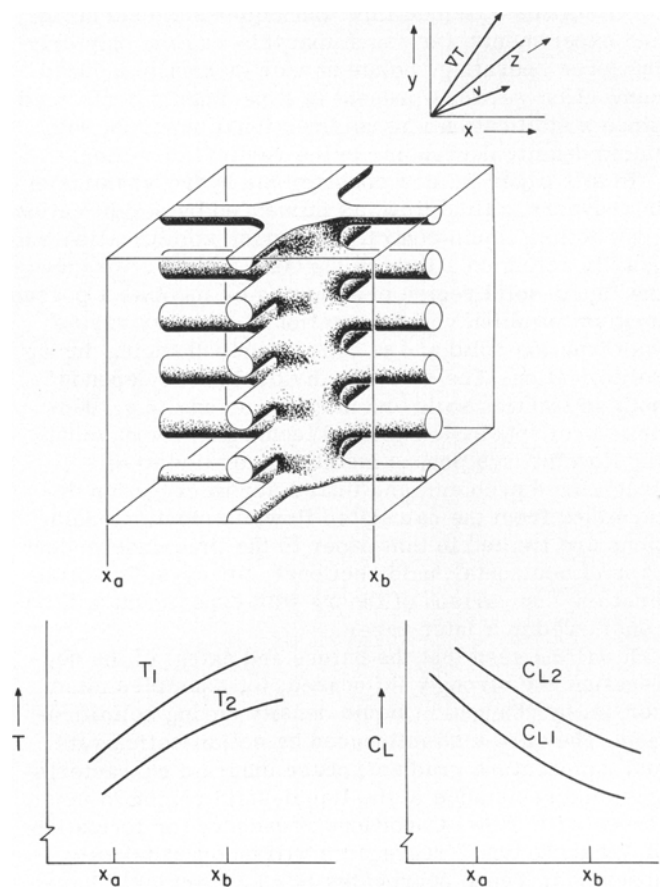

Fig. 2—Schematic illustration of a three-dimensional "volume element" of a binary alloy during solidification. Isotherms are moving in the direction given by the vector ∇T (top insert). Temperature and liquid composition distribution at two successive times along the x-axis are given at the lower left and the lower right, respectively.

$$\mathbf{v} = -\frac{K}{\mu g_L}(\nabla p + \rho_L \mathbf{g}_r) \qquad [2]$$

where

μ = viscosity of interdendritic fluid, poise

K = "permeability", sq cm

p = pressure, dynes per sq cm

$\mathbf{g}_r$ = acceleration due to gravity, cm per sq sec

ρ_L = liquid density, g per cu cm

The permeability, K, is a measure of the resistance of the medium to flow. It is a second-order tensor in the general case, depending on pore size and geometry, and varying with direction. Evaluation of K is discussed further below.

Now, when flow takes place through the volume element, conservation of total mass in the element gives (Eq. [5], Ref. 1):

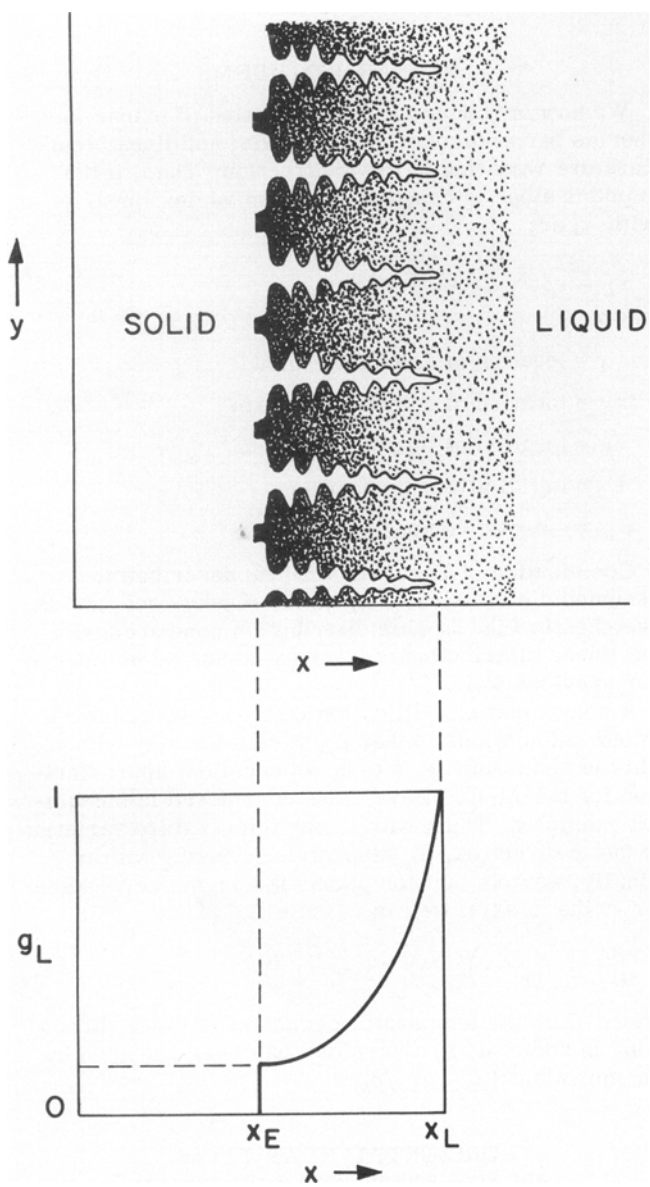

Fig. 3—Solidification model, planar isotherms. Top: sketch of the liquid-solid region. Bottom: distribution of fraction liquid in the liquid-solid region.

$$\frac{\partial \bar{\rho}}{\partial t} = -\nabla \cdot \rho_L g_L \mathbf{v} \qquad [3]$$

where:

t = time, sec

$\bar{\rho}$ = local average density (of liquid and solid) g per cu cm

The change in total mass of the element is the sum of the changes of the liquid and solid masses: (Eq. [12], Ref. 1):

$$\frac{\partial \bar{\rho}}{\partial t} = \frac{\partial}{\partial t}(\rho_s g_s + \rho_L g_L) \qquad [4]$$

where:

ρ_s = solid density, g per cu cm

g_s = volume fraction solid

And for constant solid density, assumed also in deriving Eq. [1]:

$$\frac{\partial \bar{\rho}}{\partial t} = \frac{\partial g_L}{\partial t}(\rho_L - \rho_s) + g_L \frac{\partial \rho_L}{\partial t} \qquad [5]$$

Now, ρ_L is a function only of C_L and hence of temperature, T, in the liquid-solid zone. T is a function only of x, y, z and time, t. Thus, from the chain rule:

$$\nabla \rho_L = \frac{1}{\epsilon}\frac{\partial \rho_L}{\partial t}\nabla T \qquad [6]$$

Combining Eqs. [2] to [6] now provides an expression for pressure distribution in the mushy zone during solidification:

$$\left[\nabla \cdot \frac{\rho_L K}{\mu}(\nabla p + \rho_L \mathbf{g}_r)\right] \cdot \frac{\nabla T}{\epsilon} = (\rho_L - \rho_s)\frac{\partial g_L}{\partial t}\frac{\nabla T}{\epsilon} + g_L \nabla \rho_L \qquad [7]$$

RELATION OF PERMEABILITY TO STRUCTURE

In the remainder of this paper, we now assume the permeability, K, is independent of orientation and is, therefore, a scalar. We recognize that this may not always be a valid approximation. As an example, it seems reasonable in columnar grains that flow channels parallel to the grain axes should be different in character than those perpendicular to these axes. To directly relate the permeability, K, of a porous medium to its internal structure would require solution of the fluid flow problem on a microscopic scale; and this has not been done. Approximate relations between K and geometric variables have, however, been derived by approximating flow behavior by one or another simple model.[7-9] The model we employ here is one of flow through a simple bundle of capillary tubes.

Imagine the "volume element" described above to be composed of a number of dendrite arms with one or more flow channels through each interdendritic space in the x direction. All flow channels are first imagined as being of approximately circular cross section and of equal radius. Now a permeability test is performed in which liquid is caused to flow unidirectionally through the volume element in the x direction under a pressure gradient $\partial p/\partial x$. The flow velocity through each circular tube is related to pressure gradient by

the familiar equation (assuming the component of $\mathbf{g}_r$ in the x direction is zero):

$$v_x = -\frac{r^2}{8\mu}\left(\frac{\partial p}{\partial x}\right) \quad [8]$$

Now, taking n as the number of flow channels per unit area along the face perpendicular to the x direction:

$$g_L = n\pi r^2 \quad [9]$$

Next, adding a geometric factor, c, to account for the fact that the channels are neither straight, nor perfectly symmetrical:

$$v_x = -\frac{g_L}{8\mu n\pi c}\left(\frac{\partial p}{\partial x}\right) \quad [10]$$

In the more general three-dimensional model, we assume that flow can take place in three orthogonal directions, as opposed to the one-dimensional model above. Then, Eq. [1] becomes:

$$\mathbf{v} = -\frac{g_L}{24\mu n\pi c^3}(\nabla p + \rho_L \mathbf{g}_r) \quad [11]$$

Comparison of Eq. [11] with Eq. [2] now shows that, for scalar K:

$$K = \gamma g_L^2 \quad [12]$$

where:

$$\gamma = \frac{1}{24 n\pi c^3} \quad [13]$$

For a given dendrite arm spacing we assume n and c to be constants, and, therefore, γ is constant. Thus, the permeability, K, (for a given dendrite arm spacing) is directly proportional to the square of the fraction liquid as given by Eq. [12].

CALCULATION OF INTERDENDRITIC PRESSURES AND VELOCITIES

Eqs. [1], [2], [7], and [12] now relate interdendritic pressure, p, interdendritic velocity, $\mathbf{v}$, fraction liquid, g_L, and liquid composition C_L. The equations can be solved simultaneously to yield these variables as a function of position. Numerical solution is, however, greatly simplified by making a first approximation (also made in our previous papers) that separates the fluid flow portion of the problem from the solute redistribution portion. This approximation is, that for purposes of flow calculation only, g_L is assumed to depend only on T. Then, from the chain rule,

$$\nabla g_L = \frac{1}{\epsilon}\frac{\partial g_L}{\partial t}\nabla T \quad [14]$$

And Eqs. [2], [7], and [14] can now be solved simultaneously to yield p and $\mathbf{v}$ as a function of position without aid of Eq. [1].

In our previous papers[1-4] we assumed, for flow calculations, a linear variation of g_L with position in the mushy zone. Here we use the better approximation that the distribution of fraction liquid vs temperature (g_L vs T) is that which pertains at steady state in the absence of macrosegregation. This distribution is readily calculated from equations presented earlier, by assuming steady-state, unidirectional heat and fluid flow. Derivation of this steady-state distribution is given in Appendix A, and with this approximation of g_L vs T, Eq. [7] can be solved directly, with appropriate boundary conditions, to determine $\mathbf{v}$ as a function of position and time (or liquid composition) in a given solidifying ingot.

CALCULATION OF MACROSEGREGATION

With local interdendritic velocities $\mathbf{v}$ known as a function of position and liquid composition, macrosegregation is readily calculated from the "local solute redistribution equation", Eq. [1], recalling that β is a variable equal to $(\rho_S - \rho_L)/\rho_S$. Integration of Eq. [1] is as previously described to determine the final local average composition, $\bar{C}_S$ (and, hence, macrosegregation) according to the relation:

$$\bar{C}_S = \frac{\rho_S \int_0^{1-g_E} C_S \partial g_S + \rho_{SE}\, g_E C_E}{\rho_S(1-g_E) + \rho_{SE} g_E} \quad [15]$$

where g_E = volume fraction eutectic. Eq. [15] has been previously given as Eq. [25], Ref. 1.

PLANAR ISOTHERMS

We now introduce the simplification of planar isotherms perpendicular to the x axis, and linear temperature variation in the x direction. Then, if the liquidus slope is constant, C_L also varies linearly with x, or:

$$\frac{x - x_E}{x_L - x_E} = \frac{C_L - C_E}{C_0 - C_E} \quad [16]$$

where:

x = location of volume element

x_E = location of eutectic isotherm

x_L = location of liquidus isotherm

C_0 = initial alloy composition

C_E = eutectic composition

Combination of Eqs. [26] and [16] describes the assumed distribution of g_L in the mushy zone, and is sketched in Fig. 3. This distribution compares with the linear distribution of g_L vs x, assumed in most of our previous work.[1-4]

A convenient simplification that we have employed in our calculations is that ρ_L varies linearly with C_L. This is shown in Fig. 4 to be an excellent approximation for the Al-Cu alloys to be considered in numerical examples. Thus, with linear temperature variation in the x direction, ρ_L also varies linearly with x. Finally, we note that for planar isotherms perpendicular to the x axis, we can rewrite Eq. [1] as

$$\frac{\partial g_L}{\partial C_L} = -\left(\frac{1}{1-k}\right)\left(\frac{\rho_L}{\rho_S}\right)\left[1 + \frac{v_x G}{\epsilon}\right]\frac{g_L}{C_L} \quad [17]$$

where G is the temperature gradient in the x direction. In addition, β, a variable, has been replaced by its equivalent $(\rho_S - \rho_L)/\rho_S$.

HORIZONTAL HEAT FLOW, STEADY-STATE SOLIDIFICATION

For simple numerical examples of solution of macrosegregation equations, we consider now the ingot of

Fig. 5 solidifying unidirectionally with horizontal heat flow so gravity acts in the y direction. This type of solidification can readily be carried out in the laboratory and is of practical importance in that it is analogous to solidification from the side walls of a large ingot.

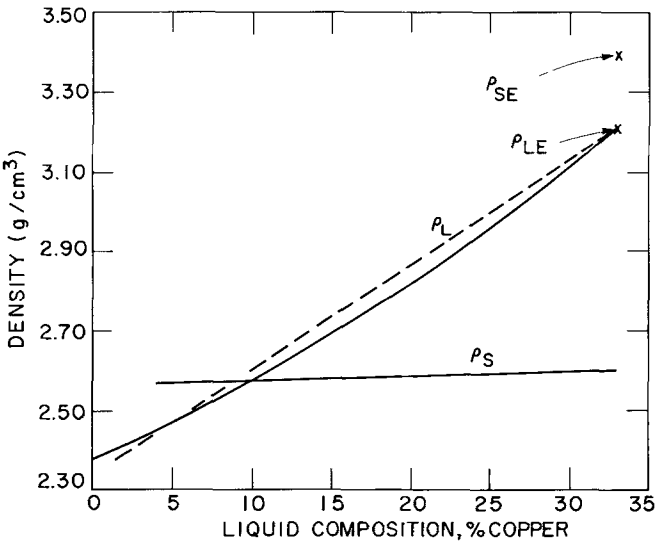

Fig. 4—Liquid and solid densities for Al-Cu alloys. Solid density is plotted vs the composition of the liquid with which it is in equilibrium in the liquid-solid region. Dashed line is liquid density assumed in calculations. Crosses are assumed eutectic liquid and solid densities, ρ_{LE} and ρ_{sE}, respectively. Solid density used in calculations is 2.62 (g per cu cm). Solid lines represent data of Elwood and Silcock[12] and Gebhardt.[13]

Boundary conditions are shown schematically in Fig. 5(c). Velocity in the y direction, v_y, is zero at the impermeable upper and lower walls, and also zero at the free surface $x = x_L$ (where $\partial p/\partial y$ is exactly $\rho_L \mathbf{g}_r$). At the eutectic isotherm, the flow component in the x direction must be just sufficient to feed eutectic shrinkage or, from continuity:

$$(v_x)_{x=x_E} = -\frac{\rho_{sE} - \rho_{LE}}{\rho_{LE}} U_E \qquad [18]$$

where U_E = velocity of the eutectic isotherm.

Since gravity is in the y direction, Eqs. [2] and [13] may be combined and rewritten as:

$$v_x = -\frac{\gamma g_L}{\mu} \frac{\partial p}{\partial x} \qquad [19a]$$

$$v_y = -\frac{\gamma g_L}{\mu}\left(\frac{\partial p}{\partial y} + \rho_L \mathbf{g}_r\right) \qquad [19b]$$

Now for the calculations to follow, we assume the chill face is below the eutectic temperature, and G/ϵ is constant (*i.e.*, solidification is "steady state"). Thus, all isotherms, including the eutectic isotherm, are moving at the same velocity, $-\epsilon/G$. Further, since it is steady state, there can be no compositional variation in the x direction. There is a net flow of interdendritic liquid from right to left to feed shrinkage, and there is also local flow in the y direction with the heavier solute-rich liquid near the dendrite roots tending to settle. Settling of this solute-rich liquid results in compositional differences in the y direction, with higher copper concentrations in the lower part of the ingot.

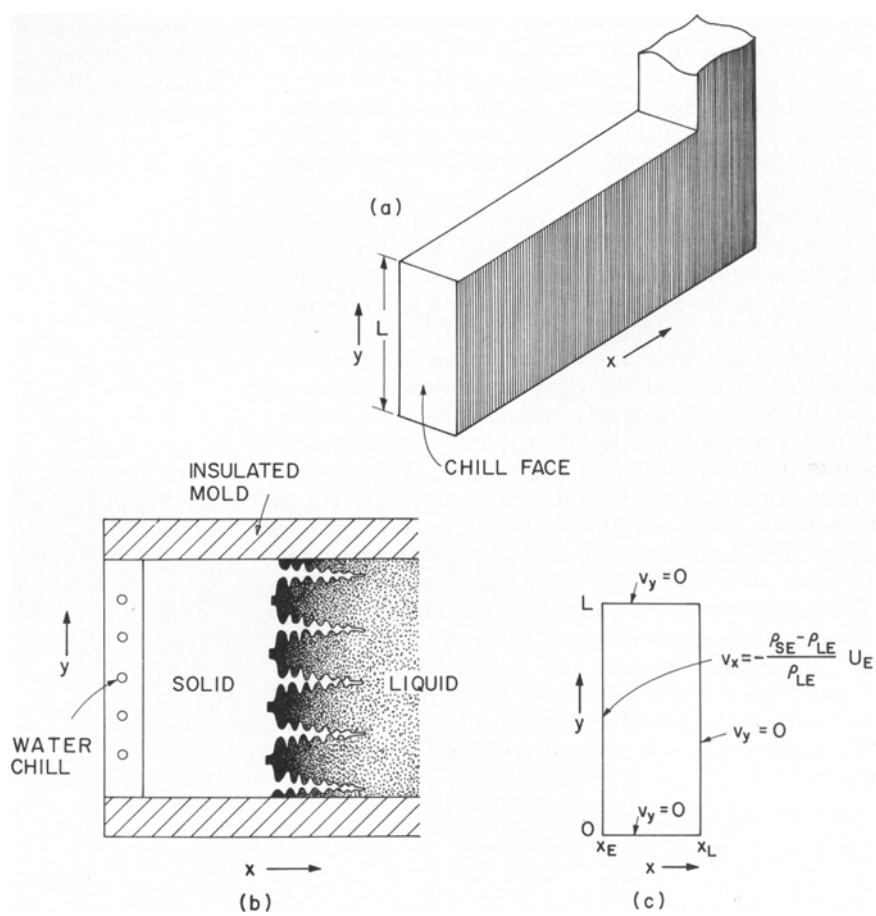

Fig. 5—Unidirectionally solidified horizontal ingot: (a) Sketch of ingot, (b) sketch of mushy zone, (c) boundary conditions for interdendritic flow.

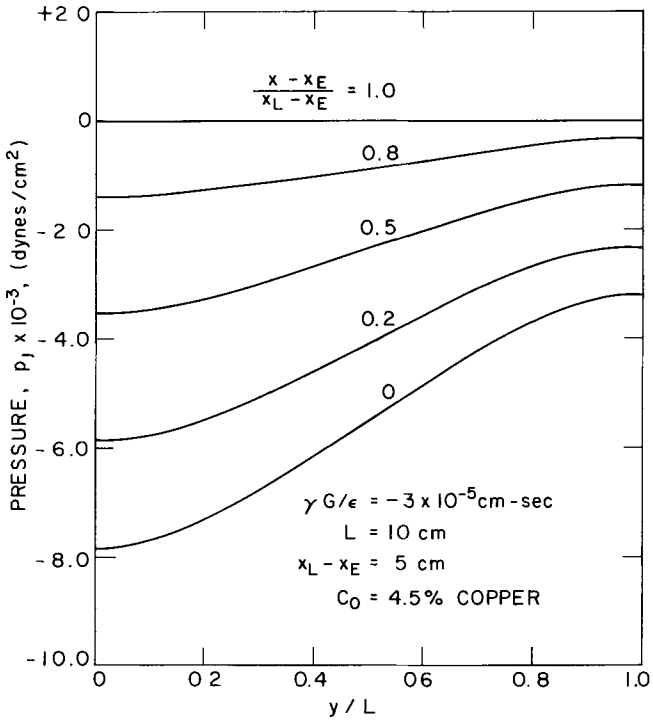

Fig. 6—Pressure, p_j, vs fractional distance along the y axis in the mushy zone.

RESULTS OF CALCULATIONS FOR HORIZONTAL HEAT FLOW, STEADY-STATE SOLIDIFICATION

In this section we present results of calculations for the mold configuration and solidification conditions of Fig. 5. The solution for velocity and pressure distributions is obtained using Eqs. [7] and [19] with the boundary conditions of Fig. 5. The local solute redistribution Eq. [17] and solute summation expression [15] are then used to calculate final macrosegregation.

Inspection of the basic differential equations and boundary conditions shows that flow behavior and segregation are a function, not of ϵ, G, or γ independently but of the parameter $\gamma G/\epsilon$, and a series of the plots to follow are for a value of $\gamma G/\epsilon = -3 \times 10^{-5}$ cm-sec. This value was chosen because it yields a moderate amount of segregation in Al-4.5 pct Cu alloy in a 10-cm high ingot with mushy zone 5 cm wide. This segregation is roughly that observed in experiments to be reported in a later paper. Most of the figures to follow are for an ingot of this size (10 cm high with a 5-cm mushy zone). The value of viscosity, μ, for Al-4.5 pct Cu is taken as 3×10^{-2} poise, and all other data are contained in Fig. 4 or in the equilibrium phase diagram for the Al-Cu binary system. Solution of the equations was by direct numerical methods using a digital computer.

Figs. 6 and 7 plot pressures in the mushy zone of the horizontally solidifying ingot of Al-4.5 pct Cu alloy. For convenience, the pressures plotted are p_i vs x where

$$p_i = p - p_0 - \rho_{L0} g_r (L - y) \qquad [20a]$$

and p_j vs y where

$$p_j = p - p_0 - \rho_L g_r (L - y) \qquad [20b]$$

where L is ingot height, p_0 is atmospheric pressure, and ρ_{L0} is liquid density of C_0 composition. Differentiation of Eqs. [20] and substitution in Eqs. [19] shows that the interdendritic velocities in the x and y directions, respectively, are proportional to the slopes of these curves:

$$v_x = -\frac{\gamma g_L}{\mu} \frac{\partial p_i}{\partial x} \qquad [21a]$$

$$v_y = -\frac{\gamma g_L}{\mu} \frac{\partial p_j}{\partial x} \qquad [21b]$$

Note from Fig. 6 that there is a finite gradient of p_j in the y direction throughout most of the mushy zone, and therefore, there is flow in this direction (settling of the relatively dense solute rich liquid). The pressure gradient is zero only at the upper and lower surfaces where boundary conditions require no flow in the y direction.

There is a gradient of p_i vs x throughout the mushy zone and, therefore, there is flow in the x direction throughout the zone. In most of the zone, flow is from right to left; but at the extreme bottom of the ingot, the pressure gradient is reversed, and so flow is in the opposite direction.

The nature of flow in the mushy zone is perhaps best seen by flow lines, and these are shown in Fig. 8 for the same ingot. Flow lines enter horizontally from the right and bend down gradually as a result of gravity. Flow reversal is seen to occur at the lower right. Flow lines are drawn to show both flow direction and velocity. Flow velocity is approximately inversely proportional to line spacing.

The only component of flow that is necessary for segregation calculation is that in the x direction, and Fig. 9 plots $v_x G/\epsilon$ vs x for several different values of y/L. It will be seen later that there is no resulting segregation at the midpoint, $y/L = 0.5$. The curve of $v_x G/\epsilon$ vs position in the mushy zone is just that given analytically for steady-state solidification, Eqs. [16],

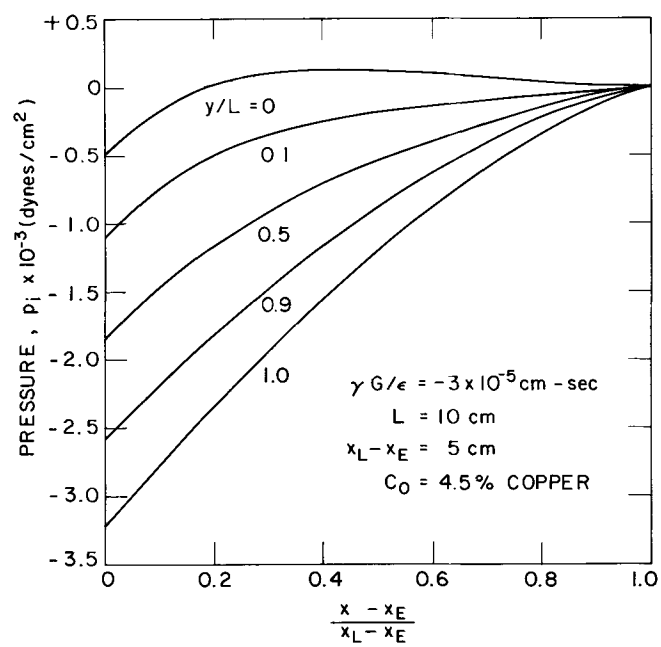

Fig. 7—Pressure, p_i, vs fractional distance along the x axis in the mushy zone.

[25], and [26]. Negative segregation results for locations (in the y direction) at which the curve for $v_x G/\epsilon$ vs x lies above the steady-state curve. Positive segregation results when the reverse is true.

An additional point of interest from Fig. 9 is that positive flow velocities, v_x, are small fractions of velocities of isotherms $-\epsilon/G$. For example, consider the lower right corner where the flow vectors, in Fig. 8, point from left to right. $v_x G/\epsilon$ here is -0.03, and therefore, flow velocity in the x direction is less than $\frac{1}{30}$ that of isotherm velocity. Thus, although flow vectors point from left to right, in this case no fluid actually leaves the moving mushy zone.

Macrosegregation is now readily calculated from curves such as that of Fig. 9, using Eqs. [15], [16], and [17]. Fig. 10 shows a plot of this segregation vs y/L for the same ingot considered above. Segregation is negative at the top (–0.55 pct Cu), and positive at the bottom (+0.65 pct Cu). Macrosegregation is zero at the mid-plane.

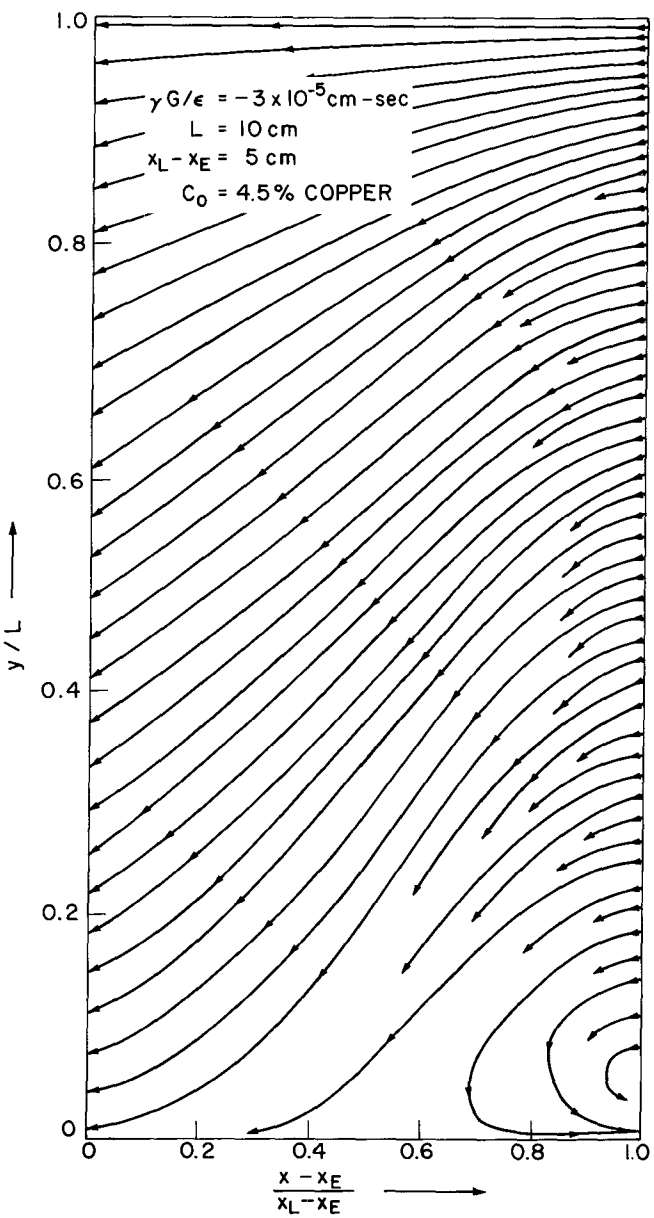

Fig. 8—Interdendritic flow lines. Flow velocity approximately inversely proportional to line spacing.

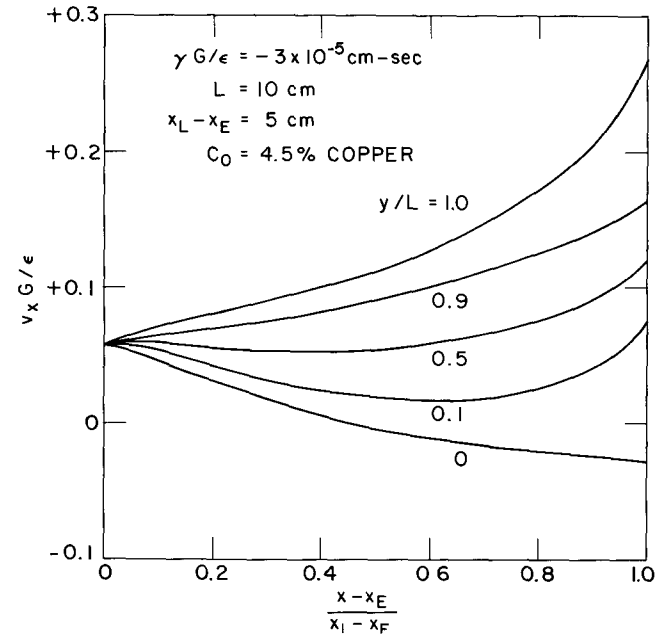

Fig. 9—($v_x G/\epsilon$) vs fractional distance along the x-axis in the mushy zone.

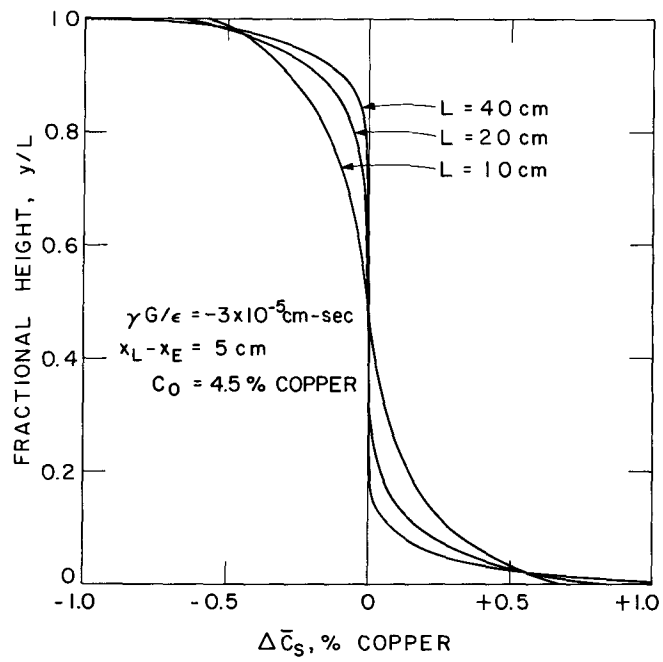

Fig. 10—Effect of ingot height on macrosegregation.

Also shown in Fig. 10 is the effect on macrosegregation of increasing ingot height with constant mushy zone thickness. As the ingot height becomes larger, differences in metallostatic head within the mushy zone become larger, and circulation of flow becomes more pronounced. Near the central portion of these ingots, however, segregation is reduced. That this result is reasonable may be seen by visualizing an extremely tall ingot, horizontally solidified at steady state. Central portions of the ingot, where flow is not significantly influenced by the upper and lower boundary conditions, must solidify as "steady state" with no macrosegregation. Thus, the taller the ingot, the

greater should be the portion of the ingot near C_0. Macrosegregation at the upper and lower boundaries should, however, be enhanced in the taller ingots from the greater metallostatic head in these ingots.

Fig. 11 shows that decreasing width of the mushy zone $(x_L - x_E)$ has an effect qualitatively similar to that of increasing ingot height. It enhances segregation at the upper and lower surfaces while decreasing segregation in the central portions of the ingot.

The parameter $\gamma G/\epsilon$ must, of course, have a marked effect on segregation since this parameter describes, for a given alloy, the interdendritic resistance to flow and the time available for flow. Fig. 12 shows this relation. Final segregation increases with increasing absolute values of $\gamma G/\epsilon$.

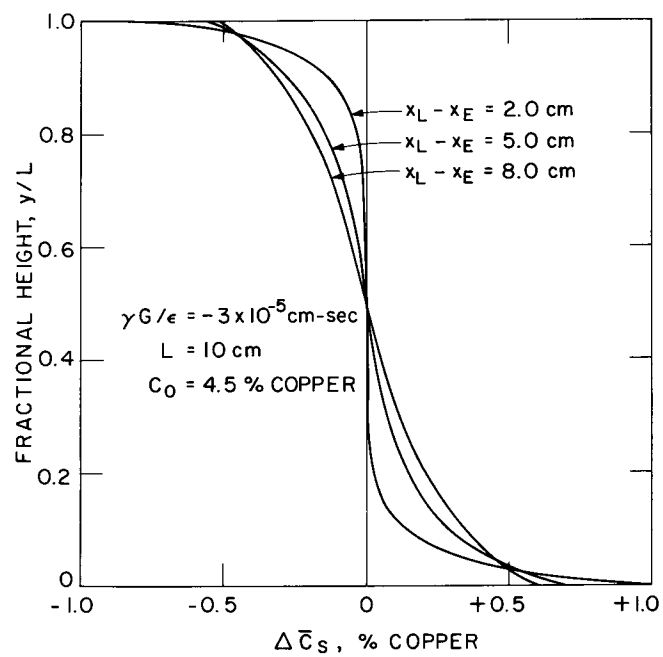

Fig. 11—Effect of mushy zone thickness on macrosegregation.

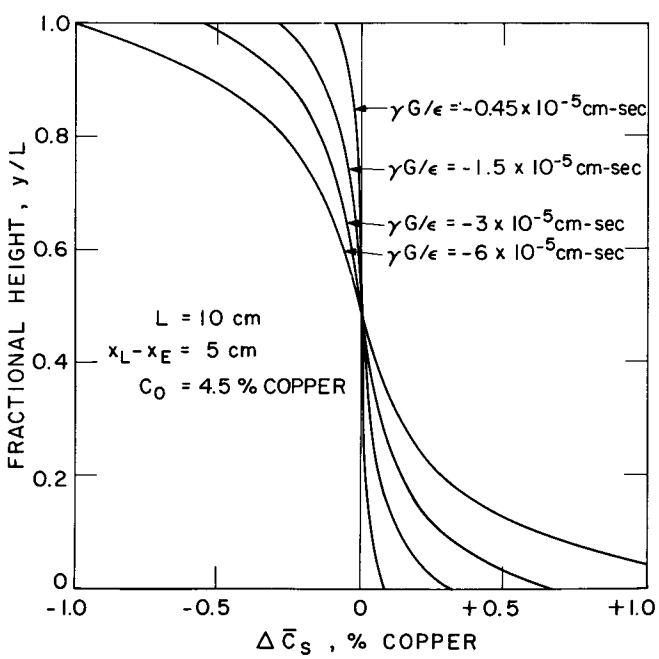

Fig. 12—Effect of $\gamma G/\epsilon$ on macrosegregation.

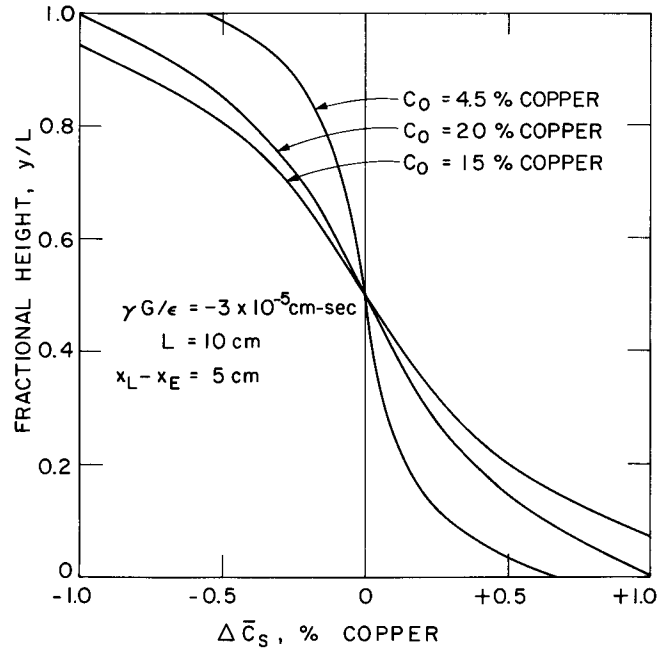

Fig. 13—Effect of initial alloy composition on macrosegregation.

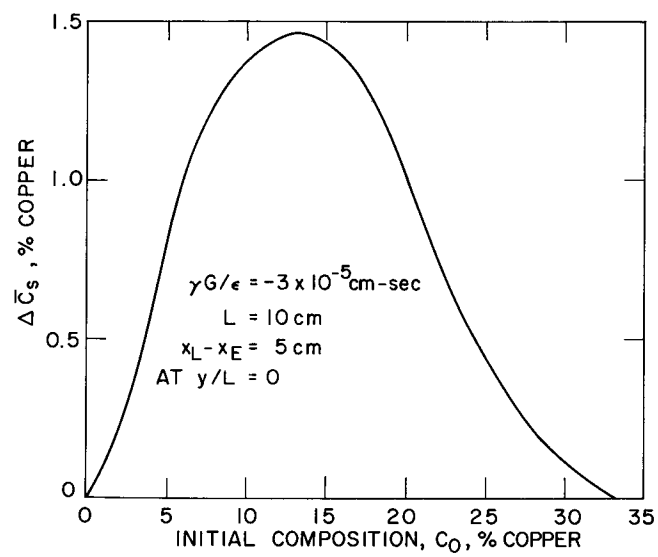

Fig. 14—Macrosegregation vs initial alloy composition at the bottom of the ingot.

Segregation must be zero for the pure alloy and the eutectic, and so must be a maximum at some composition in between. Fig. 13 shows segregation for several alloys in the ingot and Fig. 14 is a plot of maximum segregation (at the ingot base) vs copper content. The maximum lies between 10 and 15 pct Cu as it does in the case of simple inverse segregation.

FLOW INSTABILITY

Up to this point we have assumed the number of effective flow channels, n, remains constant during solidification, and that flow velocity through each channel in a given volume element is uniform. Here

we examine conditions necessary to maintain this "stable" flow and show that under certain flow conditions, flow may become unstable. Local remelting then results, with consequent formation of concentrated "channels" of highly segregated liquid.

Fig. 15 is a schematic illustration of flow channels in a volume element under conditions of "stable" flow, "intermediate" flow, and "unstable" flow. Stable flow occurs whenever $(\mathbf{v} \cdot \nabla T/\epsilon) > 0$; that is, whenever flow is from hotter to cooler regions. This can most easily be seen by first multiplying Eq. [1] by $(\partial C_L/\partial T)_{x,y,z}$ to obtain the local fraction of liquid as a function of temperature.

$$\frac{\partial g_L}{\partial T} = -\left[\frac{1-\beta}{1-k} \cdot \frac{g_L}{C_L} \cdot \frac{\partial C_L}{\partial T}\right]\left(1 + \frac{\mathbf{v} \cdot \nabla T}{\epsilon}\right) \qquad [22]$$

The quantity inside the brackets is always negative, regardless of the value of k. The larger the value of $\mathbf{v} \cdot \nabla T/\epsilon$, the more positive is $\partial g_L/\partial T$ and, hence, the more negative is $\partial g_s/\partial T$; that is, the greater is the increase of fraction solid in the channel with decrease in temperature.

Now, imagine a perturbation such that one of the channels in the volume element sketched in Fig. 15 becomes slightly larger than the surrounding channels. Resistance to flow through this channel is decreased, and flow (absolute value of $\mathbf{v}$) is increased, whatever its direction. If, however, flow vector is in a direction of decreasing temperature $[(\mathbf{v} \cdot \nabla T/\epsilon) > 0]$, the faster flow results in faster solidification in the larger channel, and the size of that channel decreases. Thus, the stable configuration is one of uniform size channels, since a driving force is present to counteract small scale perturbations. We refer to this case as "stable" flow.

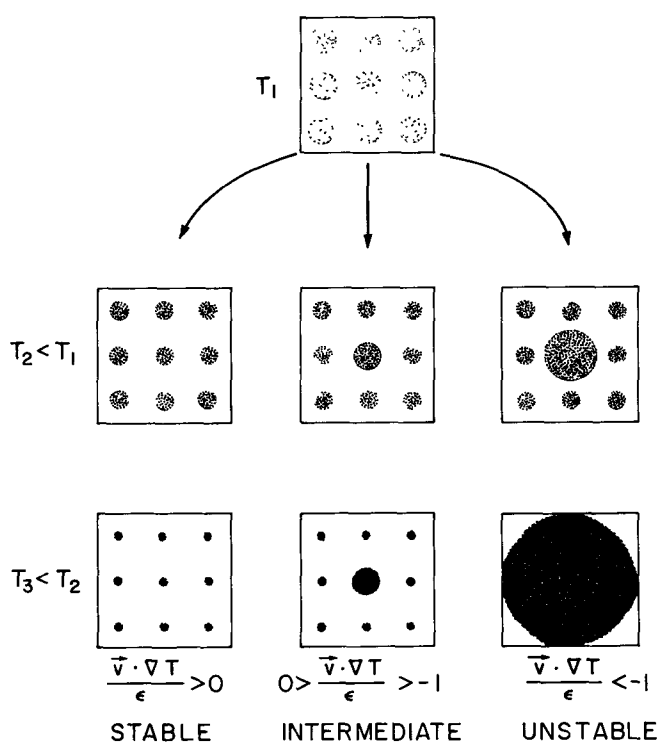

Fig. 15—Schematic illustration of change in flow channel behavior during cooling of a volume element.

Next, consider the effect of a minor perturbation in the size of a flow channel when some flow occurs in the direction of isotherm movement. Specifically, consider the case when $\mathbf{v} \cdot \nabla T/\epsilon$ is less than zero but greater than -1. From Eq. [22], the greater the flow, the less is the rate of solidification in a given channel. As sketched in Fig. 15, therefore, the size of the perturbed channel does not decrease as rapidly as that of its neighbors during solidification. However, as long as $\mathbf{v} \cdot \nabla T/\epsilon$ remains within the limits given above, $\partial g_L/\partial T$ remains positive, and so no remelting occurs.

Once, however, the condition $(\mathbf{v} \cdot \nabla T/\epsilon) < -1$ is reached in the perturbed channel, remelting does occur here, and the size of the perturbed channel actually grows during cooling of the volume element. Growth can occur, we believe, until it becomes optically visible as a "freckle" or other "channel-type" segregate.

In the ingot considered in detail in the previous section, $\mathbf{v} \cdot \nabla T/\epsilon$ (*i.e.*, $v_x G/\epsilon$) was greater than zero at all locations except the extreme base, see Figs. 8 and 9. Hence, "channel-type" segregates would be impossible here. At the extreme base, $v_x G/\epsilon$ was as low as -0.03, and so flow here would be classed as just barely "intermediate". "Channel-type" segregates are not expected here although we cannot positively rule out the possibility since our equations do not strictly apply for "intermediate" flow. In "intermediate" flow, the local flow rate through a slightly perturbed hole would gradually increase relative to the flow rate in neighboring small holes. Thus, especially when the calculated value of $v_x G/\epsilon$ is close to -1, flow in a perturbed hole might well become "unstable" as solidification progresses.

Freckles ("channel-type" segregates) would be far more likely for the ingot whose calculated interdendritic flow behavior is sketched in Fig. 16. Comparison of this figure with Fig. 8 shows much more flow in the positive direction. In this ingot, the calculated value of $v_x G/\epsilon$ drops to -2.9 at the base of the ingot, and flow is thus well into the "unstable" region. The flow lines of Fig. 16 are a much oversimplified view of the flow behavior, since actual flow must be strongly influenced by the "channeling" expected to occur.

SEGREGATION IN COMMERCIAL INGOTS

In previous sections of this paper, we have dealt primarily with segregation in ingots of simple geometry unidirectionally solidified (horizontally) under steady-state conditions. However, the basic differential equations derived, Eqs. [1] to [15], are quite general and applicable to the much more complicated heat and fluid flow conditions of commercial ingots. In this section, we consider briefly qualitative application of the concepts of this paper to segregation in such ingots. The alloy considered in the following discussion is one with a partition ratio, k, less than unity, and one which shrinks on solidification.

Consider, first of all, centerline segregation in static cast ingots. Near the end of solidification, the liquid-solid zone is as shown schematically in Fig. 17(*a*) (for an ingot solidifying with planar isotherms). For a fluid of constant density, hydrodynamics dictate the flow lines point slightly outwards as sketched in Fig. 17(*b*). The lines would turn in this direction

since 1) the boundary conditions at the walls require a flow component in the x direction (outwards), and 2) restriction to fluid flow is less near the centerline where fraction solid is lower. If the flow lines point outwards to sufficient extent, segregation will be negative near (but not at) the centerline. This "negative" segregation will be particularly severe near the base of the liquid-solid zone (with flow lines as are drawn in Fig. 17(*b*)), and we have previously proposed such flow as the cause of the negative "cone" of segregation at the base of large ingots.

If, now, liquid density is no longer constant but increases with decreasing temperature, a driving force exists to reduce the outward curvature of flow lines drawn in Fig. 17(*b*), or to turn them inwards as in Fig. 17(*c*). Segregation is therefore increased (positively). If the flow lines are turned inward to sufficient extent ($v_x G/\epsilon < -1$) channel segregates form ("freckles", "V" segregates, "A" segregates).

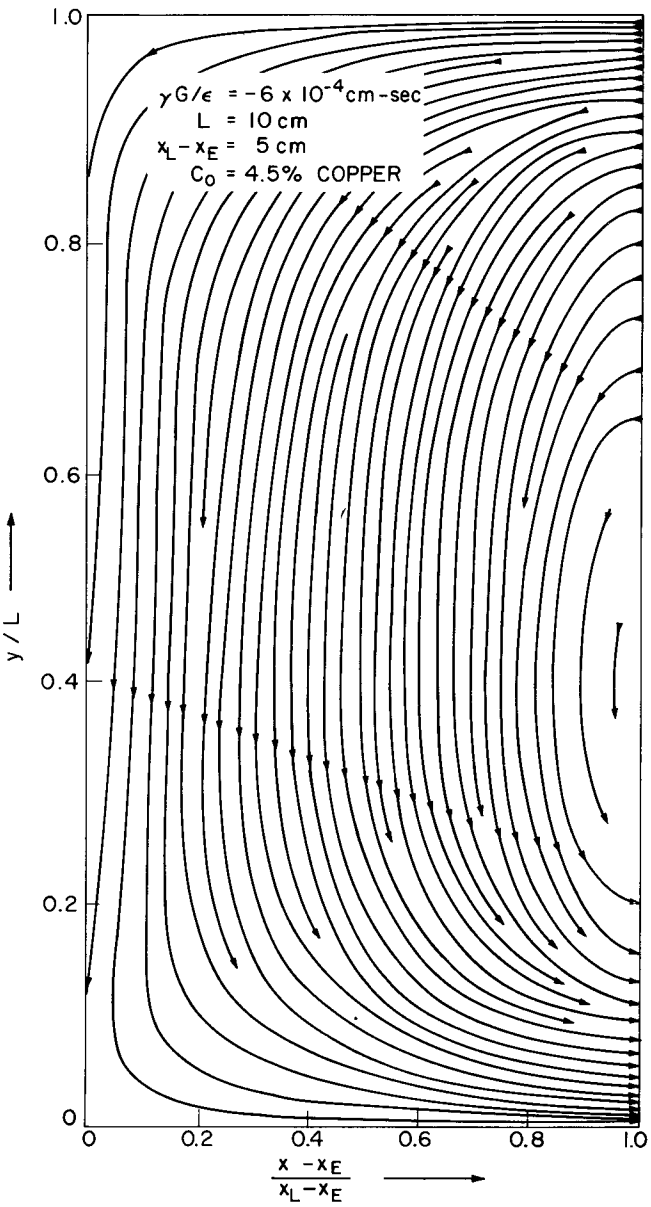

Fig. 16—Interdendritic flow lines. Flow velocity approximately inversely proportional to line spacing.

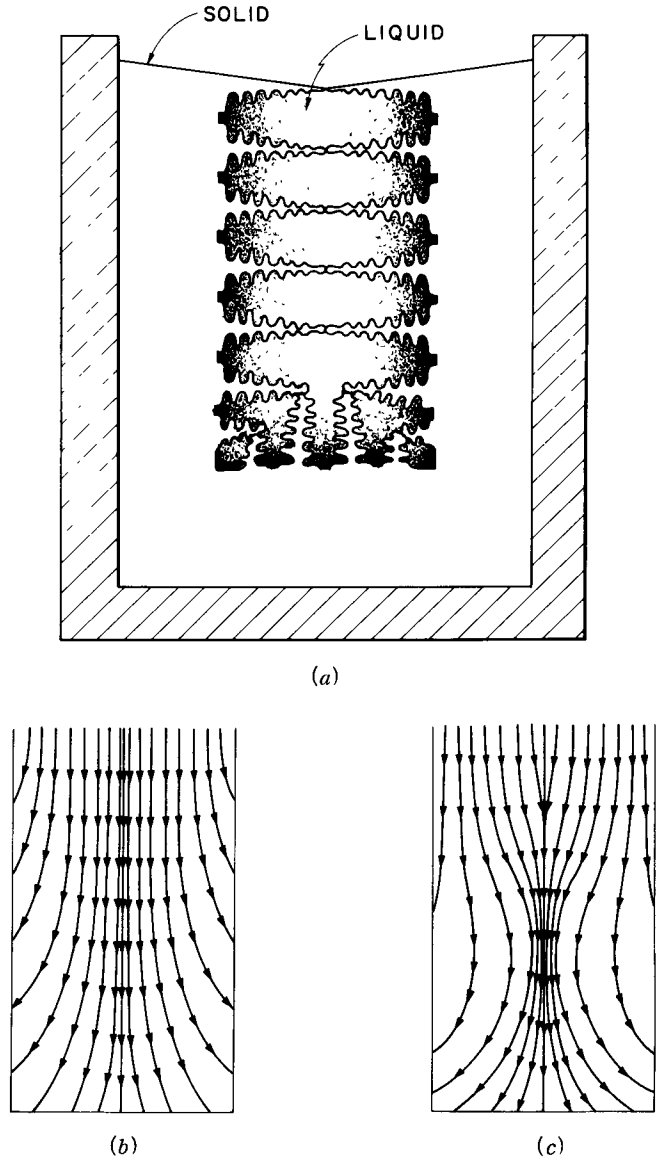

Fig. 17—Formation of centerline segregation: (*a*) sketch of the liquid-solid region at the time of interest, after the liquidus fronts have met at the centerline; (*b*) flow lines expected if fluid is of constant density; (*c*) flow lines expected if fluid density increases with decreasing temperature in the liquid-solid region.

"Freckles" and "A" segregates may, of course, also form relatively early in solidification, before liquidus isotherms from the mold walls meet. In this case they would form as in the horizontal unidirectionally solidified ingot discussed in the previous section.

It seems likely that the simple flow instability mechanism discussed above is not always the sole cause of "V" segregates. A second possible mechanism is one in which "V" shaped fissures first form by "internal hot tearing" (as a result of solidification contractions and/or atmospheric pressure acting on the solidifying mushy zone). After formation of such fissures, flow along them would be enhanced. Fissures or portions of fissures in which the critical velocity is reached ($v_x G/\epsilon < -1$) would grow as above. Flow lines would again be as in Fig. 17(*c*), their direction being strongly influenced by the effect of the "internal hot tearing" on the local magnitude and directionality

of the permeability, K. Fig. 18 shows schematically how "V" type channel segregates would be expected to form with the same general orientation as flow lines.

Fig. 19 is a simple sketch showing extension of these ideas to continuous or consumable electrode casting. If flow lines are vertical, and heat flow steady state, solidification is steady state, and no segregation results. Hydrodynamic considerations do not, of course, permit this simple a flow pattern. If they fan outward, as they would if liquid density is constant, segregation becomes negative. If they fan inward, as they can do as a result of either fissuring or liquid density that increases during solidification, positive segregation can result. In the case of fissuring or of sufficiently large driving force from density differences, "channel-type" segregates ("V" segregates, "freckles") are expected.

Throughout this paper, we have considered only alloys in which the density of the interdendritic liquid increases during solidification. Gravity would influence quite differently the flow in alloys in which the liquid density decreased during solidification. For example, consider flow in the simple case of horizontal unidirectional solidification discussed previously. Consider an imaginary alloy similar to Al-4.5 pct Cu alloy in all respects except that the sign of the liquid density change during solidification is reversed. Now the flow lines, Figs. 8 and 16, will be exactly the same, except inverted. The upper part of the ingot will now be segregated positively, and the lower, negatively. "Channel-type" segregates, when they form, will be in the upper portion of the ingot, pointing upward and inward.

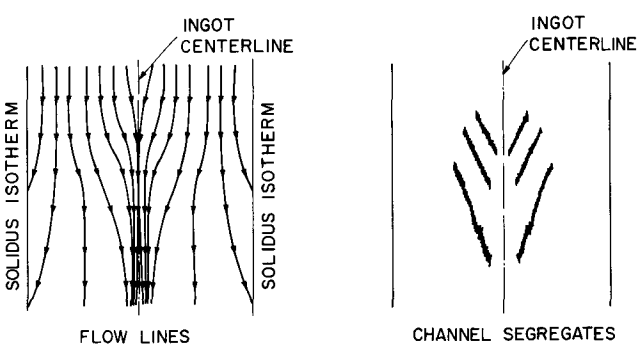

Fig. 18—Interdendritic flow lines in the liquid-solid region near the end of solidification (left), leading to channel segregates (right).

"Channel-type" segregates have been observed in various alloys in which the liquid density decreases, or is assumed to decrease, during solidification. Such alloys include hypereutectic H_2O-NH_4Cl alloys[5,6] and certain superalloys.[6] As would be expected from the foregoing, the orientation of the channel segregates in the ingot apparently depends on whether the liquid density increases or decreases during solidification.[11] However, the criterion for the formation of such "channel-type" segregates is not the sign of the density change, but the resulting value of $\mathbf{v} \cdot \nabla T/\epsilon$.

Copley and Giamei[6] have reported in detail on formation of "channel-type" segregates (freckles) in unidirectional, vertical, solidification of alloys in which liquid density decreases on solidification. Freckling is most pronounced when isotherms are most curved and location of the freckles depends on whether the curvature is convex or concave. Both observations are to be expected from the hydrodynamics discussed herein.

CONCLUSIONS

1) Principles of flow through porous media have been applied to calculation of interdendritic flow behavior during ingot solidification. Conditions are described in which flow and resulting macrosegregation are markedly affected. Numerical solutions are presented for horizontal, unidirectional, and steady-state solidification, using aluminum-copper alloys as examples.

2) "Channel-type" segregates (*e.g.*, "freckles", "A" segregates) result from a flow instability which occurs at a critical condition of flow. This condition of flow is $\mathbf{v} \cdot \nabla T/\epsilon < -1$ where $\mathbf{v}$ is the interdendritic flow velocity, ∇T is the temperature gradient, and ϵ is the rate of temperature change at a given position in the ingot.

ACKNOWLEDGMENT

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APPENDIX A

DERIVATION OF STEADY-STATE DISTRIBUTION OF FRACTION SOLID VS LIQUID COMPOSITION

Consider an ingot solidifying unidirectionally in the x direction with gravity effect negligible. Then all flow

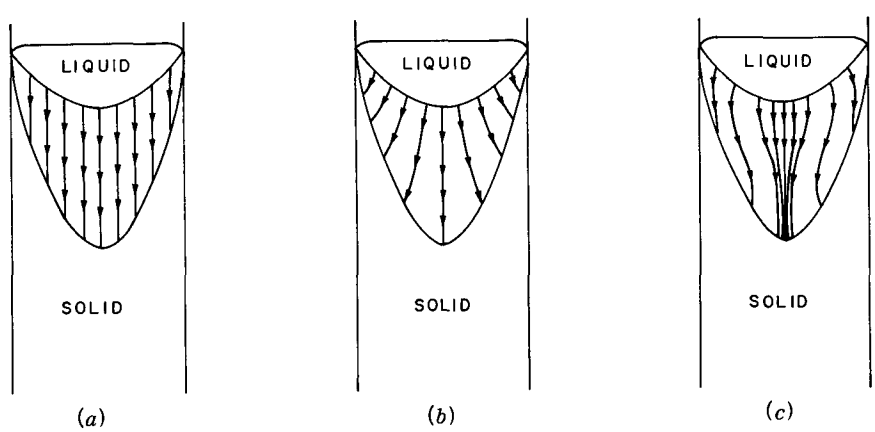

Fig. 19—Interdendritic fluid flow in consumable electrode or continuous casting: (*a*) Limiting case, all flow vertical; no segregation results, (*b*) flow resulting in negative segregation, (*c*) flow resulting in positive segregation.

is in the x direction, and combining Eqs. [2], [7], and [14] yields

$$\frac{\partial}{\partial x}(\rho_L g_L v_x) = -\frac{\epsilon}{G}\left[(\rho_L - \rho_s)\frac{\partial g_L}{\partial x} + g_L \frac{\partial \rho_L}{\partial x}\right] \qquad [23]$$

Integration over the width of the mushy zone gives

$$\rho_L g_L \frac{v_x G}{\epsilon}\Bigg|_{\rho_L g_L \frac{v_x G}{\epsilon}}^{\rho_L g_L \frac{v_x G}{\epsilon}} = -\rho_L g_L \Bigg|_{\rho_L g_E}^{\rho_L g_L} + \rho_s g_L \Bigg|_{g_E}^{g_L} \qquad [24]$$

and

$$\frac{v_x G}{\epsilon} = \frac{g_L(\rho_s - \rho_L) + g_E(\rho_{sE} - \rho_s)}{\rho_L g_L} \qquad [25]$$

Substitution of Eq. [25] in Eq. [17] and integration from $g_L = 1$ at $C_L = C_0$ to $g_L = g_E$ at $C_L = C_E$ (assuming constant k) yields the final equation for g_L as a function of C_L

$$g_L = \frac{C_L^{-1/(1-k)} + \left(\frac{\rho_s}{\rho_{sE}} - 1\right) C_E^{-1/(1-k)}}{C_0^{-1/(1-k)} + \left(\frac{\rho_s}{\rho_{sE}} - 1\right) C_E^{-1/(1-k)}} \qquad [26]$$

REFERENCES

1. M. C. Flemings and G. E. Nereo: *Trans. TMS-AIME,* 1967, vol. 239, pp. 1449-61.
2. M. C. Flemings, R. Mehrabian, and G. E. Nereo: *Trans. TMS-AIME,* 1968, vol. 242, pp. 41-49.
3. M. C. Flemings and G. E. Nereo: *Trans. TMS-AIME,* 1968, vol. 242, pp. 50-55.
4. R. Mehrabian and M. C. Flemings: *Trans. Met. Trans.*, 1970, vol. 1, pp. 455-64.
5. R. J. McDonald and J. D. Hunt: *Trans. TMS-AIME,* 1969, vol. 245, pp. 1993-97.
6. S. M. Copley, A. F. Giamei, S. M. Johnson, and M. F. Hornbecker: *Met. Trans.*, 1970, vol. 1, in press.
7. M. Muskat: *Flow of Homogeneous Fluids through Porous Media,* McGraw-Hill, New York, 1937; J. W. Edwards, Inc., Ann Arbor, 1946.
8. A. E. Scherdegger: *The Physics of Flow through Porous Media,* University of Toronto Press, 1957.
9. R. E. Collins: *Flow of Fluids through Porous Materials,* Reinhold Publishing Co., 1961.
10. E Scheil: *Metallforschung:* 1947, vol. 2, p. 69.
11. J. Preston: International Transactions Vacuum Metallurgy Conference, E. L. Foster, ed., American Vacuum Society, New York, 1968.
12. E. C. Ellwood and J. M. Silcock: *J. Inst. Metals,* 1948, vol. 74, p. 457.
13. V. E. Gebhardt, M. Becker, and S. Dorner: *Aluminum,* 1955, vol. 31, p. 315.